

On the Universal Structure of Maximal Planar Graphs

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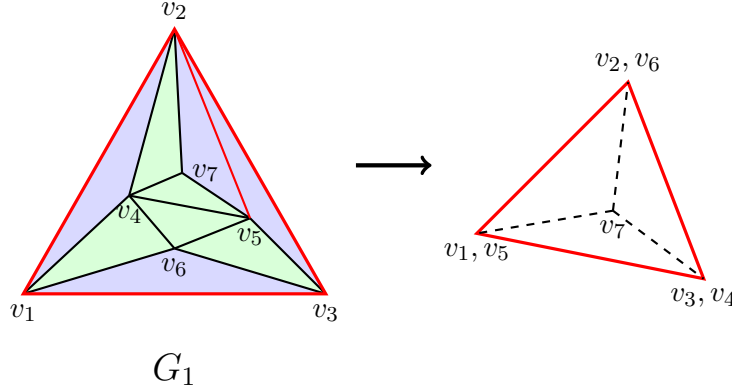
Abstract

This paper proves that for any maximal planar graph G of order n , if G contains a maximal outerplanar graph of order n as a subgraph, then there exists a homomorphism from G to K_4 . Furthermore, this class of four-colorable maximal graphs constitutes the fundamental structure of all maximal graphs, thereby allowing any maximal planar graph to be recursively decomposed into this particular type of maximal planar graph.

1

First, we illustrate the overall structure of this work with a concrete example. We then prove that it holds for all natural numbers.

As shown in the figures below.



G_1 is divided into two parts, marked by blue and green shading. These two parts each having the same number of vertices 7, edges 11, and faces 6.

Consider the green subgraph in G_1 . Clearly, it can be **folded** into K_4 . And the number of ways to fold it is 2^4 . We will prove that one of these ways of folding defines a homomorphism from G_1 to K_4 .

This process involves only two difficulties. First, whether the red edges in G_1 may collapse to a single vertex. Second, assuming the first condition is satisfied, whether there exists a vertex that cannot be folded into K_4 .

When folding a K_3 along one of its edges, the third vertex always has two possible choices. **Regarding the existence of the red edges in G_1 as constraints. We will prove that if the green subgraph in G_1 cannot be folded into K_4 then there is a K_5 exists in G_1 which contradicts the fact that G_1 is a planar graph.** This resolves the two difficulties mentioned above.

Furthermore we shall prove that every maximal planar graph admits a decomposition into smaller maximal planar graphs via the structure described above, analogous to the decomposition of solvable groups.

2

We now generalize the conclusion of the first part to all planar graphs. We begin by recursively defining the maximal outer planar graphs considered in the first part. To do this, we need to use the concept of “gluing” from topology.

Definition 1. Let $\{u_i\}$ be a set whose elements are K_3 ’s. G is called a maximal outer planar graph, denoted by $\overline{R}_n$. Define as follows

1. $\overline{R}_0$ is u_0 .
2. $\overline{R}_1$ is obtained by gluing a K_2 in u_0 and a K_2 in u_1 together from the outside.
3. $\overline{R}_n$ is obtained by gluing a K_2 that hasn’t been glued yet in u_i , $i < n$ and a K_2 in u_n together from the outside.

Definition 2. Normal Graph

For any maximal planar graph G , if G contains a subgraph $\overline{R}_n$ with the same number of vertices as G , then G is called a normal graph denoted by $\overline{S}_n$. In particular, if every K_3 face in G has no vertices in its interior, then G is called a minimal normal graph.

Theorem 1. If every K_3 face of a maximal planar graph G has no interior vertices, then G is a minimal normal graph.

Proof. All K_3 ’s in G form a finite sequence $\{u_k\}$. One can find a subsequence $\{u_{k_n}\}$ such that the $\overline{R}_n$ formed by $\{u_{k_n}\}$ contains all vertices of G .

□

From the definitions, S_n and $\overline{R}_n$ each have $n + 3$ vertices, $2n + 3$ edges, and $n + 2$ faces. We have

Theorem 2. *For any maximal planar graph G whose order is bigger than 3, there exists a minimal normal subgraph $\overline{S}_n$ such that $n > 0$ and $\overline{S}_n$ contains the three outermost vertices of G .*

Proof. We denote by $V'(G)$ the set consisting of all vertices of G **except its three outermost vertices**. Since G is a maximal planar graph, for any one of its edges, there exist several vertices in $V'(G)$ such that each of these vertices is adjacent to the two endpoints of that edge, thereby forming a sequence of K_3 's that are nested by inclusion.

Keep the largest one of this sequence of K_3 's and delete all vertices and edges in its interior. If this vertex is connected to all three outermost vertices, then the resulting subgraph is $\overline{S}_1$.

Apply the above procedure repeatedly until no small K_3 inside the outermost K_3 contains any interior vertex. The graph obtained in this way is the $\overline{S}_n$ we are looking for.

□

By Theorem 1, every maximal planar graph admits a composition series analogous to that of a solvable group.

Definition 3. *Let G be a maximal planar graph. $\overline{S}_n$ is the minimal normal subgraph of G . Module of G , denoted by $G^{(j)}$ is defined as follows*

1. $G^{(0)}$ is $\overline{S}_n$.
2. $G^{(j)}$ is the union of $G^{(j-1)}$ and all minimal $\overline{S}_n$ corresponding to the subgraphs associated with each K_3 in $G^{(j-1)}$.

We have

Theorem 3. *G is a maximal planar graph if and only if there exists a composition series as follows.*

$$\overline{S}_n = G^{(0)} \triangleright G^{(1)} \triangleright G^{(2)} \triangleright \dots \triangleright G^{(j)} = G$$

Proof. The sufficiency has already been established in the above definition; we only prove the necessity.

By Definition 3, each newly added part in the composition series is a union of several $\overline{S}_n$, where the outermost K_3 of each $\overline{S}_n$ is exactly a K_3 in some $\overline{S}_n$ from the previous term. Hence, G is planar.

□

We are now ready to prove the main theorem of this paper.

Theorem 4. *For any $\overline{S}_n$, there exists a homomorphism from $\overline{S}_n$ to K_4 .*

Proof. Assume that for all 2^n ways of folding $\overline{R}_n$, there always exists a vertex that cannot be folded into K_4 due to the restriction of $\overline{S}_n$. We shall show that this assumption implies that $\overline{S}_n$ has a K_5 -minor.

Consider an arbitrary folding $\phi : \overline{R}_n \rightarrow K_4$. Suppose that a vertex v cannot be folded into K_4 under ϕ . For every vertex x in K_4 , there exists a set X such that X consists of all preimages of x that are adjacent to v . Since each vertex has two choices during the folding process, some elements in X under ϕ can still be moved to other vertices of K_4 without creating conflicts. We denote by X' the set consisting of those vertices that **cannot** be moved under ϕ . By the assumption that none of the 2^n folding schemes defines a homomorphism from $\overline{S}_n$ to K_4 . Sets X'_1, X'_2, X'_3 and X'_4 have the following properties:

1. X'_i is **nonempty**.
2. **Every element of X'_i is adjacent to at least one element in each of other three sets.**

Delete all vertices of G except those belonging to X'_i ($i = 1, 2, 3, 4$). Then perform edge contractions to the resulting subgraph until each X'_i is reduced to a single vertex. This yields a K_4 -minor. Since each of these four vertices is adjacent to v , we obtain a K_5 -minor. This contradicts the fact that G is planar.

Thus, among the 2^n ways of folding $\overline{R_n}$, at least one admits a homomorphism from $\overline{S_n}$ to K_4 .

□

Combining Theorem 3 and Theorem 4, it follows that every maximal planar graph is 4-colorable.

3

We now state the direct condition underlying the above proof. This condition is not that G has no K_5 -minor, but that G is planar.

Definition 4. *as follows*

A graph G is said to admit a crossing-free embedding into $\mathbb{R}^{m-2}$ if there exists a mapping of G into $\mathbb{R}^{m-2}$ such that for any $(m-2)$ -polytope realized by a subgraph H , no edge of G crosses it.

Using Definition 3, we can easily define maximal graphs in $\mathbb{R}^{m-2}$.

Definition 5. *as follows*

A graph G is called a maximal graph if adding any edge to G destroys the existence of a crossing-free embedding of G into $\mathbb{R}^{m-2}$.

All definitions and theorems in Section 2 can be extend directly to $\mathbb{R}^{m-2}$. We have

Theorem 5. *Every graph that admits a crossing-free embedding into $\mathbb{R}^{m-2}$ is m -colorable.*

References

- [1] R. Diestel, *Graph Theory*, 5th ed., Springer, Berlin, 2017.
- [2] D. B. West, *Introduction to Graph Theory*, 2nd ed., Prentice Hall, 2001.